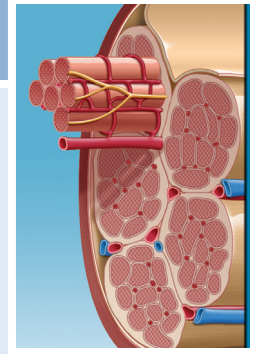


Basic Science of Graft Tissue in Sports Medicine

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Graft tissue is widely used in sports medicine for soft tissue reconstruction and augmentation of ligaments and tendons. Understanding the physiology of fixation and incorporation of tendon and/or bone within the tunnels is important for the surgeon, because this knowledge can help guide postoperative rehabilitation and educate patients when they return to play. Many options exist for graft tissue, both as autografts and allografts, and for the processing of these grafts. The healing of such grafts has been most carefully studied in patients who have undergone anterior cruciate ligament (ACL) reconstruction, for whom a significantly increased rate of graft failure with allografts has been found clinically.^{1,2}

AUTOGRAFT

Investigators have conducted numerous studies to examine graft incorporation and viability of autografts using the patellar tendon (PT) and semitendinosus (ST) in animal models. These grafts have been examined both individually and in comparison.

BONE-TENDON-BONE AUTOGRAFT

Several studies of patellar tendon autografts have been performed. A rabbit model of patellar tendon autografts by Amiel et al. demonstrated that PT autografts undergo a process of “ligamentization” and that cells responsible for this process were of extra-graft origin.³ The investigators came to this conclusion because they found that the autografts were centrally acellular with a peripheral rim of cells at 2 weeks, progressed to having a central focal proliferation of cells at 3 weeks, and then progressed to cellular homogeneity by 4 weeks after surgery. Microscopically, the patellar tendon autografts gradually changed from having the properties of the patellar tendon to having those seen in the native ACL. By 30 weeks after surgery, an increase in type III collagen and glycosaminoglycan content was observed from the levels typically found in PTs to those found in native ACLs, and collagen-reducible cross-link analysis demonstrated that grafted tissue changed from the normal PT pattern of low dihydroxy-lysine norleucine and high histidinohydroxymerodesmosine to the opposite pattern seen in the normal ACL.³

PT autograft healing was also studied in a goat model at 3 and 6 weeks postoperatively. The total anteroposterior (AP) translation significantly increased from 3 to 6 weeks, with in situ forces in the graft decreasing by as much as 22.2% at 6

weeks. At 3 weeks, the mode of failure was the graft pulling out of the tibial tunnel, which changed to a mix of midsubstance failures at 6 weeks. Histologic evaluations revealed progressive and complete incorporation of the bone block in the femoral tunnel but only partial incorporation of the tendinous part of the graft in the tibial tunnel. The investigators did not study any longer time points in this study.⁴

Lastly, in a rabbit model using medial one-third PT autografts, investigators concluded that on gross inspection the autografts did not resemble the control ACLs. Biomechanically, AP knee laxity was more than double that of the control knees at 52 weeks, and the strength of the PT autografts plateaued at 30 weeks, but the ultimate load to failure and stiffness never reached those of the control native ACLs. Histologically, the autografts progressed from being hypercellular with a random collagen fiber bundle organization to having a near normal cellularity with a more parallel collagen fiber bundle pattern.⁵

AUGMENTED AUTOGRAFT

In an attempt to augment the healing of bone-patellar tendon-bone (BPTB) autograft, Yasuda et al.⁶ performed ACL reconstruction in the bilateral knees of 20 canines. The left knee either had a fibrin sealant with transforming growth factor- β and epidermal growth factor or fibrin sealant alone. The right knee served as a control, with no augmentation. These investigators found that the application of the growth factors increased the stiffness and maximum failure load of the femur-graft-tibia complex at 12 weeks, with no difference between the fibrin sealant alone and the control knees. A qualitative histologic analysis showed that most of the cells in the grafts treated with growth factors had spindle-shaped nuclei, whereas cells in the other grafts had round-shaped nuclei. The investigators concluded that these growth factors improved the structural properties.⁶

To examine the effect of collagen platelet composites and platelet concentration on BPTB autografts, Spindler et al.⁷ placed a collagen-platelet composite around autologous PTs and compared these with goats that received the collagen scaffold only and their contralateral native knees. At 6 weeks after surgery, the average increase in AP laxity was 40% less in the collagen-platelet composite group than in the group that had collagen wrapped around their grafts alone at 30 degrees. Significant correlations were found between serum-platelet concentration and AP laxity, maximum load to failure, and graft stiffness. The

authors concluded that use of a collagen-platelet composite with a BPTB autograft enhanced healing and that a higher serum-platelet count was inversely correlated with sagittal plane laxity and highly predictive of ACL reconstruction graft strength and stiffness at 6 weeks.⁷

SEMITENDINOSUS OR SOFT TISSUE AUTOGRAFT

Numerous studies have also been performed on ST autografts, with findings largely similar to those of the PT studies. In rabbits, bony fixation of the ST graft was complete by 26 weeks, but large differences persisted in the strength and stiffness of the graft compared with the native ACL and ST tendons at 52 weeks. By 52 weeks, the graft did not fail because of pullout from the tunnels, but rather by intrasubstance rupture.⁸ In a study by Grana et al., it was found that failure of the graft occurred in the midsubstance rather than from pullout as early as 3 weeks postoperatively. The authors found that fixation occurs by an intertwining of graft and connective tissue, with anchoring to bone by collagenous fibers and bone formation in the tunnels that have the appearance of the Sharpey fibers seen in an indirect tendon insertion.⁹ Papachristou et al.¹⁰ examined the histologic changes that occurred between 3 and 12 weeks using ST tendons that were harvested without detachment from their tibial insertion and free ST autografts. They found that the ST tendon when harvested without detachment from the tibial insertion retained viability without graft necrosis, whereas the free tendon group did have necrosis of the graft 3 weeks after surgery, with progressive revascularization at 6 and 12 weeks postoperatively.¹⁰

Studies of ST autografts have also been performed in sheep models.^{11,12} In one study it was found that the tendon graft was predominantly acellular at 2 weeks, with the core portion remaining necrotic even at 12 weeks. The area of necrosis had disappeared at 24 and 52 weeks. At all time points, the AP translation of the reconstructed knee remained significantly greater than that of the control subject, and the load to failure was still less than that of the control at 52 weeks.¹² In another study in sheep using ST autografts, the authors did not find evidence of any graft necrosis. They observed that the random collagen fiber orientation progressed to a longitudinal orientation from the peripheral to the central areas of the graft over the initial 12 weeks, with a uniform sinusoidal crimp pattern similar to that seen in the normal ACL in nearly half of each graft by 24 weeks, with further maturation at 52 weeks.¹¹

The differences in healing between ST and BPTB autografts were studied in beagles. Soft tissue grafts were placed in the left knees, and BPTB grafts were placed in the right knee. The soft tissue graft was anchored to the tunnel wall with collagen fibers resembling Sharpey fibers by 12 weeks. In the BPTB graft, the bone plug was anchored with newly formed bone at 3 weeks, although osteocytes in the plug trabeculae were necrotic for 12 weeks. In load to failure testing of the soft tissue graft, the graft failed at the graft-bone interface at 3 weeks and then at the midsubstance by 6 weeks. In the BPTB graft, the graft failed at the graft-wall interface at 3 weeks and the proximal site in the

bone plug at 6 weeks. The ultimate failure load of the soft tissue graft was significantly inferior to that of the BPTB graft at 3 weeks, but a significant difference was not observed 6 weeks postoperatively.¹³

Marumo and colleagues evaluated the process of graft “ligamentization” in humans in the first study to address biochemical properties of autografts in living subjects after ACL reconstruction.¹⁴ From a cohort of 50 patients who underwent “clinically effective” hamstring or BPTB reconstruction, biopsies of grafted tissue were collected at 4 to 6 months and 11 to 13 months postoperatively. These were compared with tissue samples from cadaveric native ACL, patellar tendon, and semitendinosus and gracilis tendons. The investigators found that the total collagen content and nonreducible/reducible crosslink ratios increased significantly during the postoperative period and closely resembled the native ACL by one year.¹⁴

ALLOGRAFT

Numerous studies also have been performed on allograft tendons in animal models. An important consideration in these studies is the processing of the allograft, which differs between the studies.

Freeze-dried BPTB allografts in a goat model were compared with contralateral controls. Allograft revascularization occurred within the first 12 weeks, and the grafts matured to resemble normal connective tissue within 26 weeks. Graft stiffness and maximum load to failure was 29% and 43% of the control values, respectively. The authors concluded that the results of freeze-dried PT allografts were biomechanically and biologically similar to the published results of PT autografts.¹⁵

Goertzen et al.¹⁶ compared deep-frozen gamma-irradiated (2.5 mrad) BPTB allografts with argon gas protection to BPTB allografts without gamma irradiation in a canine model. At 12 months, the irradiated allografts had a load to failure that was 63.8% of the contralateral normal ACLs. The nonirradiated allografts had a load to failure of 69.1% of the load to failure of the control subjects. Histologic analysis demonstrated that the allografts appeared to be developing well-oriented collagen fibers compared with the normal ACL. A modified microangiographic technique demonstrated similar vascularity to normal ACL in the nonirradiated allograft, compared with slight hypervascularization in the irradiated group.¹⁶

Using a sheep model, Zimmerman et al.¹⁷ compared frozen, untreated allografts to frozen grafts that were processed with a chloroform-methanol solvent extraction technique and another group of frozen tendons treated with a permeation-enhanced extraction technique. Two months after surgery, enhanced cellular repopulation was noted in both chemically treated allograft groups compared with the untreated grafts. Mechanical testing at 6 months after surgery showed statistically similar anterior drawer resistance in all grafted knees; however, the two chemically processed grafts had significantly decreased stiffness compared with the untreated grafts, and both treatment groups also tended to be weaker.¹⁷

Fromm et al.¹⁸ investigated the revascularization and reinnervation of cryopreserved ACL allografts in a rabbit model. They found minimal immune response to the grafts, with considerable

revascularization by the 24th postoperative week. Reinnervation was complete by 24 weeks.¹⁸

Arnoczky et al.¹⁹ compared fresh and deep-frozen PT allografts in a dog model. Deep-frozen PT allografts appeared to undergo remodeling in a manner comparable with that observed in autogenous patellar-tendon grafts, with avascular necrosis followed by revascularization and cellular proliferation. At 1 year after reconstruction, the gross and histologic appearance of the deep-frozen patellar tendon allograft resembled that of a normal ACL. The fresh PT allografts incited a marked inflammatory and rejection response depicted by perivascular cuffing and lymphocyte invasion. The investigators concluded that the deep-frozen PT grafts were more likely to survive within the knee joint than the fresh allografts as a result of the absence of an obvious rejection response and the similarity of the remodeling process.¹⁹

Harris et al.²⁰ looked at tibial tunnel enlargement with allograft ACL reconstruction in a goat model. They found significant increases in tibial tunnel size within the first 6 weeks of healing. The increased tunnel size persisted up to 36 weeks with no further remodeling. Histologic analysis showed remodeling and incorporation of the bone plug in all cases by 18 weeks with fibrous attachment within the tunnel. All allografts had progressive ligamentization with tendon-to-tunnel wall biologic fixation with dense connective tissue. These investigators concluded that the tibial tunnel enlargement did not appear to affect the ultimate incorporation of the allograft on a histologic level.²⁰

Freeze-dried bone-ACL-bone allografts were compared with control subjects that did not undergo surgery in a goat model. At 1 year after reconstruction, the reconstructed knees had a significantly greater total AP laxity than the control subjects. Stiffness was significantly less than in control subjects. The maximum load to failure was also significantly reduced in the allograft group. Histologic evaluation and microangiography of the allografts was similar to that of the native ACL.²¹

Jackson et al.²² used deoxyribonucleic acid (DNA) probe analysis to determine the fate of donor cells in fresh BPTB allografts after transplantation in a goat model. They found that donor DNA was completely replaced by recipient DNA in the transplanted ligaments within a 4-week period.²²

AUGMENTATION OF ALLOGRAFT

Because of decreased structural properties found in allografts, mesenchymal stem cells (MSCs) have been investigated as potential agents to enhance bone tunnel and tendon healing. In a study by Soon et al.,²³ bilateral ACL reconstructions were performed in a rabbit model using Achilles tendon allografts, with the graft on a single side being coated with autogenous MSCs in a fibrin glue carrier. These investigators found a significantly higher load to failure in the group treated with MSCs. Histologic analyses at 8 weeks showed mature scar tissue resembling Sharpey fibers in the control group and a mature zone of fibrocartilage blending from bone to the allograft in the MSC-treated group that resembled the normal ACL insertion.²³ Li et al.²⁴ also looked at modifying irradiated Achilles allografts in a rabbit model with autogenous MSCs or platelet-derived growth factor- β (PDGF- β) transfected MSCs. Bilateral ACL reconstructions were again

performed, with the left knee allograft being seeded with either MSCs or PDGF- β transfected MSCs. The right knee served as a control. Seeding the allograft with MSCs (auto or PDGF- β) was found to accelerate cellular infiltration and enhance collagen deposition.²⁴

Fleming et al.²⁵ studied the effect of collagen-platelet composites on BPTB allografts in a porcine model. The collagen-platelet composite group was compared with animals receiving an allograft alone. After 15 weeks of healing, the AP laxity values of the reconstructed knees and the load to failure were superior in the collagen-platelet composite group compared with the group that received the allograft alone. In addition, no regions of necrosis were found in the collagen-platelet composite group, but regions of necrosis were found in the nonaugmented grafts.²⁵

A BPTB allograft with synthetic polypropylene augmentation was compared with a fresh BPTB allograft and a cryopreserved BPTB allograft in a sheep model.²⁶ In this study, cryopreservation did not have any effect on graft characteristics. Gross and histologic examination did not reveal any significant differences between the augmented and nonaugmented groups at any of the time periods. The augmented group had significantly reduced AP translation at 52 weeks compared with the nonaugmented group. The ultimate tensile strength was significantly higher in the augmented group at 4 weeks, but at 52 weeks both groups had attained only 50% of the normal ACL strength.²⁶

HEALING OF AUTOGRAFT VERSUS ALLOGRAFT

Three studies examining the difference between allografts and autografts have also been performed. All studies found a slower biologic incorporation for allografts, with decreased load to failure at 6- and 12-month time points. Jackson et al.²⁷ used similar-sized PT autografts and fresh-frozen allografts to reconstruct the ACL in goats. These autografts and allografts were evaluated at 6 weeks and 6 months postoperatively. The investigators found that whereas the structural and material properties of autografts and allografts at time zero were similar, differences in healing occurred during the first 6 months. The allografts demonstrated a greater decrease in their structural properties, a slower rate of biologic incorporation, and the prolonged presence of inflammatory cells. At 6 months the autograft had improved stability and increased strength to failure.²⁷ Dustmann et al.²⁸ and Scheffler et al.²⁹ found similar findings in sheep models when they compared soft tissue autografts with identical non-sterilized fresh-frozen allografts.^{28,29} Revascularization and recellularization and reformation of the collagen crimp formation were significantly delayed at 6 and 12 weeks of healing compared with the autografts, but differences were less distinct at 52 weeks. At 52 weeks, the mechanical, structural, and AP laxity were worse in the allografts, but no difference was found in these qualities at early time points.^{28,29}

Nikolaou et al.³⁰ found conflicting results in a dog model comparing autografts with cryopreserved allografts. They found no evidence that the cryopreservation had any effect on healing, with no difference between autografts and allografts in terms of their load to failure at 36 weeks. Revascularization approached normal by 24 weeks in both groups, and no evidence of structural

degradation or immunological reaction was seen. These authors concluded that “a cryopreserved ACL allograft can provide the ideal material for ACL reconstruction.”³⁰

FUTURE DIRECTIONS

There has been renewed interest recently in the possibility of augmented native ACL repair. Murray and colleagues have developed a biologically enhanced acute primary repair that uses a whole blood-soaked collagen scaffold and suture stent to stabilize a provisional clot in the gap between the torn ACL ends.³¹ Pre-clinical porcine studies have demonstrated a healed ACL repair with similar mechanical properties to an ACL reconstruction and less development of posttraumatic osteoarthritis.^{32,33} The latter finding is subject to ongoing research on the underlying mechanism of cartilage protection—the complete interactions between collagen-platelet composites and the intra-articular tissues are yet unknown. The bio-enhanced repair technique has demonstrated safety in a small human feasibility study and has received Food and Drug Administration approval for a larger in-human clinical trial.³⁴

SUMMARY

The intra-articular biologic and biomechanical behavior of both autografts and allografts for ACL reconstruction has been extensively researched. Several common features have emerged. First, a “ligamentization” of all these grafts occurs over months. Second, all the new cells populating the graft are derived from the recipient. Third, the weakest link in early time points (<4 weeks) for both BPTB and soft tissue grafts are the fixation points within tunnels, whereas after 6 weeks it is the failure of the grafts within joint or “midsubstance.” Fourth, the bone heals more rapidly within the tunnels than soft tissue by “Sharpey-like fibers.” Fifth, evidence shows that growth factor, collagen-platelet composite, and stem cells augment the biologic incorporation and enhance the structural properties of the grafts. Sixth, the majority of studies comparing autografts and allografts found that allografts have slower biologic incorporation and decreased structural properties. Finally, in animal models, freeze-dried BPTB, 2.5 mrad, and cryopreservation all were successful in ACL reconstructions; however, clinical results have not confirmed these observations.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Amiel D, Kleiner JB, Roux RD, et al. The phenomenon of “ligamentization”: anterior cruciate ligament reconstruction with autogenous patellar tendon. *J Orthopaed Res*. 1986;4(2):162–172.

Level of Evidence:

Basic science

Summary:

A rabbit model of patellar tendon autografts demonstrated that patellar tendon autografts undergo a process of “ligamentization”

and that the cells responsible for this process were of extragraft origin.

Citation:

Arnoczky SP, Warren RF, Ashlock MA. Replacement of the anterior cruciate ligament using a patellar tendon allograft. An experimental study. *J Bone Joint Surg Am*. 1986;68(3):376–385.

Level of Evidence:

Basic science

Summary:

Comparison of fresh and deep-frozen patellar tendon (PT) allografts in a dog model showed that deep-frozen PT allografts appeared to undergo remodeling in a manner comparable with that observed in autogenous PT grafts, with avascular necrosis followed by revascularization and cellular proliferation. The fresh PT allografts incited a marked inflammatory and rejection response depicted by perivascular cuffing and lymphocyte invasion.

Citation:

Ballock RT, Woo SLY, Lyon RM, et al. Use of patellar tendon autograft for anterior cruciate ligament reconstruction in the rabbit—a long-term histologic and biomechanical study. *J Orthopaed Res*. 1989;7(4):474–485.

Level of Evidence:

Basic science

Summary:

This study used a rabbit model with patellar tendon (PT) autografts. The authors found that the anteroposterior knee laxity of the autografts was more than double that of the control knees at 52 weeks, and the strength of the PT autografts plateaued at 30 weeks, but the ultimate load to failure and stiffness never reached those of the control native anterior cruciate ligaments.

Citation:

Grana WA, Egle DM, Mahnken R, et al. An analysis of autograft fixation after anterior cruciate ligament reconstruction in a rabbit model. *Am J Sports Med*. 1994;22(3):344–351.

Level of Evidence:

Basic science

Summary:

This study of autografts in a rabbit model demonstrated that graft failure occurs in the midsubstance, not as a result of pullout as early as 3 weeks postoperatively, and that this fixation in the bone tunnels occurs by an intertwining of graft and connective tissue and anchoring to bone by collagenous fibers and bone formation in the tunnels that have the appearance of the Sharpey fibers seen in an indirect tendon insertion.

Citation:

Harris NL, Indelicato PA, Bloomberg MS, et al. Radiographic and histologic analysis of the tibial tunnel after allograft anterior cruciate ligament reconstruction in goats. *Am J Sports Med*. 2002;30(3):368–373.

Level of Evidence:

Basic science

Summary:

This study examined tibial tunnel enlargement with allograft anterior cruciate ligament reconstruction in a goat model. The

authors found significant increases in tibial tunnel size within the first 6 weeks of healing. The increased tunnel size persisted up to 36 weeks with no further remodeling.

Citation:

Jackson DW, Grood ES, Goldstein JD, et al. A comparison of patellar tendon autograft and allograft used for anterior cruciate ligament reconstruction in the goat model. *Am J Sports Med.* 1993;21(2):176–185.

Level of Evidence:

Basic science

Summary:

This study compared patellar tendon autografts and fresh-frozen allografts in a goat model. The structural and material properties of autografts and allografts at time zero were similar; however, the allografts demonstrated a greater decrease in their structural properties, a slower rate of biologic incorporation, and the prolonged presence of inflammatory cells. At 6 months the autograft had improved stability and increased strength to failure.

Citation:

Kaeding CC, Aros B, Pedroza A, et al. Allograft versus autograft anterior cruciate ligament reconstruction: predictors of failure from a MOON prospective longitudinal cohort. *Sports Health.* 2011;3(1):73–81.

Level of Evidence:

I

Summary:

This study of 1000 patients with 94% follow-up demonstrated that in 18-year-old patients, a failure rate of 20% occurred in patients who received allografts compared with a 6% failure rate in patients who received autografts. In 40-year-old patients, a failure rate of 3% occurred in patients who received autografts compared with a 1% failure rate in patients who received allografts. The number needed to harm from using allografts is seven for high school-aged patients.

Citation:

Marumo K, Saito M, Yamagishi T, et al. The “ligamentization” process in human anterior cruciate ligament reconstruction with autogenous patellar and hamstring tendons: a biochemical study. *Am J Sports Med.* 2005;33(8):1166–1173.

Level of Evidence:

II

Summary:

This study investigated the process of ligamentization of ACL autograft in living human specimens. They found that in a cohort of 50 subjects with either BTB or hamstring autograft, surgical biopsies at 1 year postoperatively confirmed that collagen content and nonreducible/reducible crosslink ratio more closely resembled that of the native ACL rather than patellar tendon or semitendinosus/gracilis tendon.

Citation:

Spindler KP, Murray MM, Carey JL, et al. The use of platelets to affect functional healing of an anterior cruciate ligament (ACL) autograft in a caprine ACL reconstruction model. *J Orthopaed Res.* 2009;27(5):631–638.

Level of Evidence:

Basic science

Summary:

The authors used bone-patellar tendon-bone autografts in a goat model and found that placing a collagen-platelet composite around an autologous patellar tendon resulted in an average increase in anteroposterior laxity that was 40% less in the collagen-platelet composite group than in the group that had collagen alone wrapped around their grafts.

Citation:

Tomita F, Yasuda K, Mikami S, et al. Comparisons of intraosseous graft healing between the doubled flexor tendon graft and the bone-patellar tendon-bone graft in anterior cruciate ligament reconstruction. *Arthroscopy.* 2001;17(5):461–476.

Level of Evidence:

Basic science

Summary:

The differences in healing between semitendinosus and bone-patellar tendon-bone (BPTB) autografts were studied in beagles. The soft tissue graft was anchored to the tunnel wall with collagen fibers resembling Sharpey fibers by 12 weeks. In the BPTB graft, the bone plug was anchored with newly formed bone at 3 weeks, although osteocytes in the plug trabeculae were necrotic for 12 weeks. In load to failure testing of the soft tissue graft, the graft failed at the graft-bone interface at 3 weeks and then at the midsubstance by 6 weeks. In the BPTB graft, the graft failed at the graft-wall interface at 3 weeks and the proximal site in the bone plug at 6 weeks.

Citation:

Zimmerman MC, Contiliano JH, Parsons JR, et al. The biomechanics and histopathology of chemically processed patellar tendon allografts for anterior cruciate ligament replacement. *Am J Sports Med.* 1994;22(3):378–386.

Level of Evidence:

Basic science

Summary:

This study compared frozen, untreated allografts with frozen grafts that were processed with a chloroform-methanol solvent extraction technique and another group of frozen tendons treated with a permeation-enhanced extraction technique. Mechanical testing 6 months after surgery showed statistically similar anterior drawer resistance in all grafted knees; however, the two chemically processed grafts had significantly decreased stiffness compared with the untreated grafts, and both treatment groups also tended to be weaker.

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